JEDIDIAH O. THOMPSON

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EDUCATION

Stanford University — Stanford, CA
Candidate for Ph.D. in Physics — *Expected June 2023*
Advisor: Savas Dimopoulos | Stanford Institute of Theoretical Physics (SITP)

Yale University — New Haven, CT
B.S. in Physics — *June 2017*
Summa cum laude

RESEARCH INTERESTS

- Particle physics, Beyond Standard Model (BSM) physics.
- Phenomenology of ultralight dark matter candidates. Signatures in direct detection experiments, astrophysics, and cosmology.
- Novel techniques for understanding quantum field theory at strong coupling.

PUBLICATIONS

1. B. Henning, H. Murayama, F. Riva, J. O. Thompson, and M. T. Walters, *Towards a nonperturbative construction of the S-matrix*, arXiv:2209.14306.
2. D. Cyncynates, O. Simon, J. O. Thompson, and Z. J. Weiner, *Nonperturbative structure in coupled axion sectors and implications for direct detection*, Phys. Rev. D **106** (2022) 083503 [arXiv:2208.05501].
3. D. Cyncynates, T. Giurgica-Tiron, O. Simon, and J. O. Thompson, *Resonant nonlinear pairs in the axiverse and their late-time direct and astrophysical signatures*, Phys. Rev. D **105** (2022) 055005 [arXiv:2109.09755].
4. A. Arvanitaki, S. Dimopoulos, M. Galanis, D. Racco, O. Simon, and J. O. Thompson, *Dark QED from Inflation*, J. High Energ. Phys. **2021** 106 (2021) [arXiv:2108.04823].
5. A. Arvanitaki, S. Dimopoulos, M. Galanis, L. Lehner, J. O. Thompson, and K. Van Tilburg, *Large-misalignment mechanism for the formation of compact axion structures: Signatures from the QCD axion to fuzzy dark matter*, Phys. Rev. D **101** (2020) 083014 [arXiv:1909.11665].
6. W. Goldberger, S. G. Prabhu, and J. O. Thompson, *Classical gluon and graviton radiation from the bi-adjoint scalar double copy*, Phys. Rev. D **96** (2017) 065009 [arXiv:1705.09263].
7. T. Appelquist, J. Ingoldby, M. Piai, and J. Thompson, *A spartan model for the LHC diphoton excess*, arXiv:1606.00865.

SELECTIVE PROGRAMS AND AWARDS

ARCS Scholar — 2021–2022
Selective fellowship for graduate research in science

ICTP Summer School on Particle Physics — Summer 2019
Covered fundamental aspects of, and recent developments in, phenomenology

Lindau Nobel Laureate Meeting in Physics — Summer 2019
US Young Scientist Delegate to the meeting

William R. Hewlett Fellow 2017–2021
Stanford Graduate Fellowship *Stanford University*

Wohlenberg Prize 2017
Award for an outstanding senior in the field of engineering or physical sciences *Yale University*

DeForest Pioneers Prize 2017
Award for distinguished creative achievement in physics *Yale University*

Phi Beta Kappa 2015
Academic honor society

COMMUNITY OUTREACH AND LEADERSHIP

Wonderfest Science Envoy 2019–2020
Science envoy for Wonderfest, a SF Bay Area science outreach organization
Outreach talk: "What we know (and some of what we don't) about dark matter"
Interviewed on KPOO's radio show "Let Me Touch Your Mind"

SITP Student Journal Club 2018–2019
Co-organizer

KZSU Stanford Radio 2018–2019
Hosted a weekly talk show focusing on physics, math, and the arts

WYBC Yale Radio 2016–2017
Co-hosted a weekly talk show focusing on physics and math

Yale Society of Physics Students 2014–2016
President (2016), organized series of student journal talks

CONFERENCE/SEMINAR TALKS

Harvard University particle theory family seminar Oct. 31, 2022
Highly-visible coupled axions and implications for direct detection

Massachusetts Institute of Technology informal group seminar Oct. 25, 2022
Towards a nonperturbative construction of the S-matrix

Fine Theoretical Physics Institute, Univ. of Minnesota Oct. 14, 2022
Towards a nonperturbative construction of the S-matrix

PACMAN Seminar Oct. 7, 2022
Highly-visible coupled axions with implications for direct and indirect detection

Johns Hopkins University phenomenology group Sept. 12, 2022
Towards a nonperturbative construction of the S-matrix

Univ. of Washington, INT 22-2b workshop Aug. 25, 2022
Highly-visible coupled axions and implications for direct detection

17th Patras Workshop on Axions, Wimps, and Wisps Aug. 12, 2022
Friendship in the Axiverse

CERN workshop "Nonperturbative Methods in QFT" May 31, 2022
A nonperturbative S-matrix from truncation

University of Geneva May 18, 2022
Nonlinear axions and how to observe them

University of Göttingen Feb. 7, 2022
Nonlinear axions and how to observe them

Ecole Polytechnique Fédérale de Lausanne (EPFL) Oct. 11, 2021
Friendship in the Axiverse

Stanford Institute for Theoretical Physics Seminar July 26, 2021
Dark QED from Inflation

Istituto Nazionale de Fisica Nucleare (INFN) April 23, 2020
The large misalignment mechanism: Formation and signatures of compact axion structures

Kavli Institute for Particle Astrophysics and Cosmology (KIPAC) Oct. 16, 2018
Fuzzy Dark Matter

Univ. of Calif. Los Angeles (UCLA) Nov. 29, 2017
Classical gluon and graviton radiation from the bi-adjoint scalar double copy

TEACHING EXPERIENCE

Stanford University 2020–2021
- Winter 2020, Fall 2020, Winter 2021: Teaching assistant for introductory physics discussion sections
- Fall 2021: Teaching assistant for Quantum Field Theory I
- Experience with both in-person and remote learning

Yale University Physics Department 2017
- Physics Department Teaching Assistant
- Designed procedures and helped instruct Phys 382L: Advanced Laboratory
- Experiments included a Monte Carlo analysis of the 2-dimensional Ising model, measuring the electron charge and the Boltzmann constant with shot and Johnson noise respectively, and creating superconducting tunnel junctions via evaporation deposition

Crimson Research Institute 2021–Present
- Mentored high school students one-on-one as they researched an advanced physics topic
- Developed curricula in various topics, including dark matter, cosmology, and quantum mechanics

Yale University Math Department 2014–2016
- Tutored for Math 222: Linear Algebra with Applications

Yale University Science and Quantitative Reasoning Tutor 2015–2016
- Tutored for a variety of introductory mathematics and physics courses

Freshman Scholars at Yale Summer 2016
- Taught six students introductory physics, from Newton's Laws through Lagrangian mechanics

ONEXYS Coach 2016, 2017
- Online Experiences for Yale Scholars
- Tutored students in precalculus and calculus, conducted remotely over Zoom